

MULTI-UTILITY PROTECTION SYSTEM For Agricultural Equipment

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This project is used for fire detection and then sending an SMS alert to a cellphone. The design is based on Arduino Uno. The project also includes an LCD and an LED for visual indications, and a buzzer for audio indication. It has sensory systems including flame and temperature sensors along with a relay switching device.

Agricultural equipment include farm robots, pest controls, and crops and seeds monitoring systems. This circuit generates an alarm if a fire breaks out near motors, transformers or other equipment installed in the

field. At the same time, electrical load is automatically switched off. This circuit should be installed near the equipment to be monitored.

Circuit and working

The circuit diagram of the multi-utility protection system for agricultural equipment is shown in Fig. 1. It comprises LCD (JHD162A), Arduino Uno, piezo buzzer, SIM900A GSM module, preset (10-kilo-ohm), LED, 9V battery, flame sensor, temperature sensor, 5V relay module and a few other components.

If a fire breaks out, or excessive

heat or burning flame is detected near the equipment, the system generates an alert with visual LED indication and a sound alarm. Status of the environment in real time, including temperature and flame level, is displayed on the LCD.

The LCD is used to track the temperature sensor level. If the level goes beyond safe limit, immediate action can be taken. Most of the time, workers are in the field, so they can check various sensor levels through the LCD.

GSM-based SMS alert facility is also included. It allows getting the SMS alert on a cellphone at home, even at night in case of fire. The owner can then alert the worker or incharge present in the field to take appropriate action.

Flame sensor. KY-026 flame sensor module is used here. This flame detection sensor module is quite sensitive to flame or light. Detection angle can be around 30 to 60 degrees. The module operates on 5V DC. Its analogue output pin A0 is connected to pin A5 of Arduino board.

LCD module. The 16-pin JHD162A LCD module used here can display 16 characters in two lines. It is used to display the temperature level on the first line and flame level on the second.

Arduino Uno. Arduino Uno is based on ATmega328 series micro-controller (MCU). It contains 14 digital input/output pins (of which six can be used as PWM outputs), six analogue inputs, 16MHz crystal oscillator, USB connection, power jack, ICSP header and reset button.

GSM module. GSM SIM900A module works on frequencies from 900MHz to 1800MHz. It includes RS232 interface, which allows connecting the PC and MCU with RS232 chip. Baud rate can be set in the range

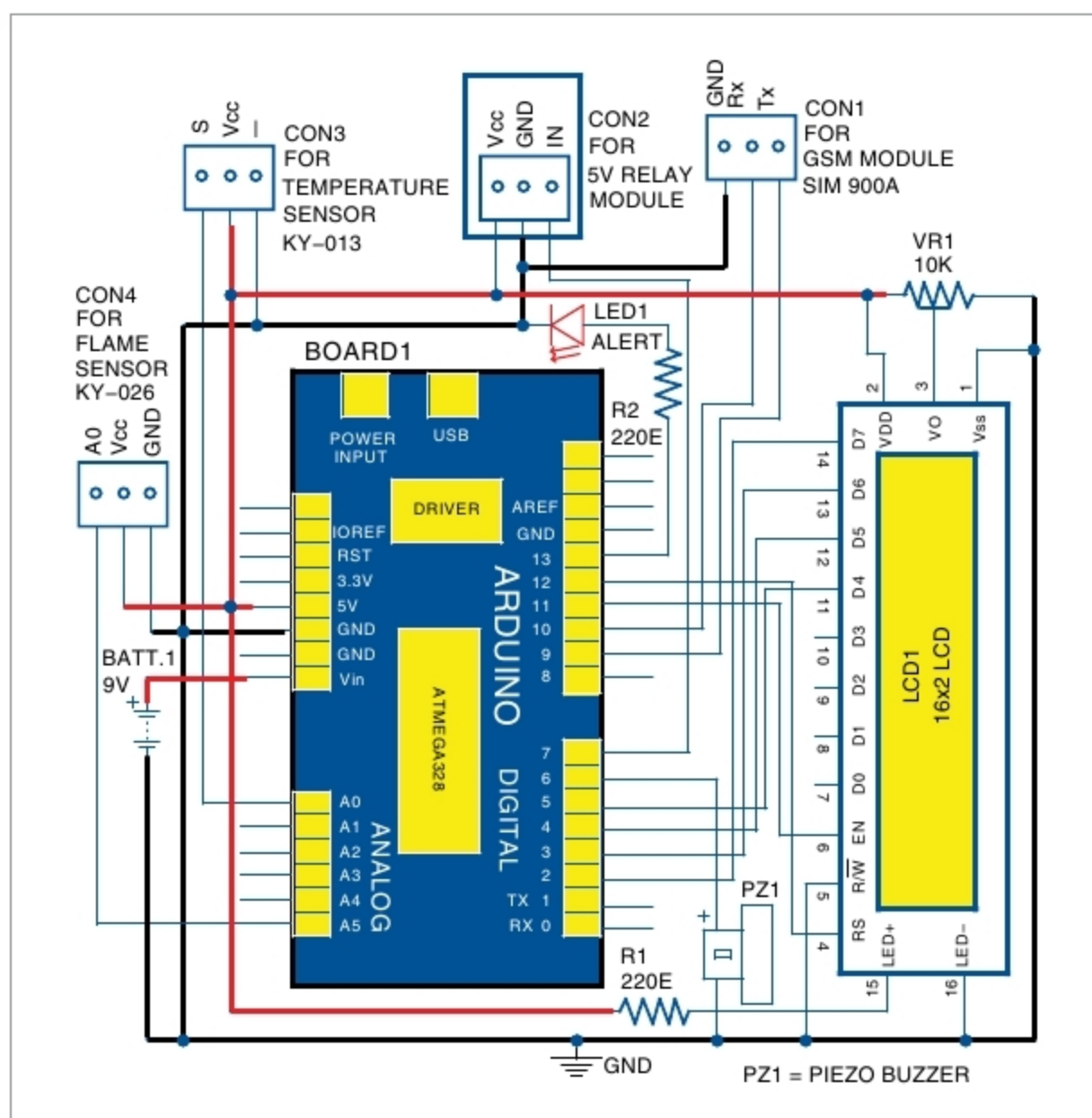


Fig. 1: Circuit diagram of multi-utility protection system